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POWER SEMICONDUCTOR MODULE HAVING POWER LEADS AND PINS, POWER ELECTRONIC SYSTEM AND METHOD FOR FABRICATING A POWER SEMICONDUCTOR MODULE

Abstract

A power semiconductor module includes: an encapsulation body having a first side, an opposite second side, and lateral sides connecting the first and second sides; a first and a second power lead exposed from a first one of the lateral sides, the first power lead configured to provide a first DC+ connection and the second power lead configured to provide a first DC- connection; a plurality of power semiconductor dies encapsulated by the encapsulation body and connected to the first and second power leads; and a first and a second pin exposed from the encapsulation body and arranged perpendicular to the first and second power leads, the first pin configured to provide a second DC+ connection and the second pin configured to provide a second DC- connection.

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Background/Summary

TECHNICAL FIELD

[0001] This disclosure relates in general to a power semiconductor module, in particular a power semiconductor module comprising power leads as well as pins. This disclosure further relates to a power electronic system comprising such a power semiconductor module as well as to a method for fabricating such a power semiconductor module.

BACKGROUND

[0002] Semiconductor modules, in particular power semiconductor modules, may be configured to operate with a high voltage of e.g. 100V or more, or 500V or more, or 1.2 kV or more, or 2 kV or more and/or a strong current of e.g. 1 A or more, or 10 A or more, or 100 A or more. Furthermore, semiconductor modules, in particular power semiconductor modules, may be configured for fast switching. Modules which comprise (power) semiconductor dies based on e.g. SiC or GaN may enable such fast switching speeds. However, with fast switching parasitic inductances may be problematic for example because parasitic inductances in the commutation loop may cause strong voltage overshoots of e.g. the drain-source current of an involved transistor due to fast commutation of the involved currents. Such an overshoot may exceed the maximum breakdown voltage of the transistor device and may cause damage.

SUMMARY

[0003] Various aspects pertain to a power electronic system, comprising: a power semiconductor module, comprising: an encapsulation body comprising a first side, an opposite second side and lateral sides connecting the first and second sides, a first and a second power lead exposed from a first one of the lateral sides, the first power lead configured to provide a first DC+ connection and the second power lead configured to provide a first DC- connection, a plurality of power semiconductor dies encapsulated by the encapsulation body and connected to the first and second power leads, a first and a second pin exposed from the encapsulation body and arranged perpendicular to the first and second power leads, the first pin configured to provide a second DC+ connection and the second pin configured to provide a second DC- connection; and a driver board arranged over the first side of the encapsulation body and comprising driver circuitry configured to drive the power semiconductor dies of the power semiconductor module, wherein the driver board further comprises a first capacitor connected to the first and second pins to provide a first commutation loop.

[0004] Various aspects pertain to a power semiconductor module, comprising: an encapsulation body comprising a first side, an opposite second side and lateral sides connecting the first and second sides; a first and a second power lead exposed from a first one of the lateral sides, the first power lead configured to provide a first DC+ connection and the second power lead configured to provide a first DC- connection; a plurality of power semiconductor dies encapsulated by the encapsulation body and connected to the first and second power leads; and a first and a second pin exposed from the encapsulation body and arranged perpendicular to the first and second power leads, the first pin configured to provide a second DC+ connection and the second pin configured to provide a second DC- connection.

[0005] Various aspects pertain to a method for fabricating a power semiconductor module, the method comprising: connecting a plurality of power semiconductor dies to a first and second power

lead, the first power lead configured to provide a first DC+ connection and the second power lead configured to provide a first DC- connection; encapsulating the plurality of power semiconductor dies with an encapsulation body, the encapsulation body comprising a first side, an opposite second side and lateral sides connecting the first and second sides, such that the first and second power leads are exposed from a first one of the lateral sides; and providing a first and second pin exposed from the encapsulation body and arranged perpendicular to the first and second power leads, the first pin configured to provide a second DC+ connection and the second pin configured to provide a second DC- connection.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] The accompanying drawings illustrate examples and together with the description serve to explain principles of the disclosure. Other examples and many of the intended advantages of the disclosure will be readily appreciated in view of the following detailed description. The elements of the drawings are not necessarily to scale relative to each other. Identical reference numerals designate corresponding similar parts.

[0007] FIGS. 1A and 1B show a top down view (FIG. 1A) and a side view (FIG. 1B) of a power semiconductor module which comprises power leads and which additionally comprises pins, both of which providing respective DC+ and DC- connections.

[0008] FIGS. 2A and 2B show a perspective view of a power electronic system (FIG. 2A) and a top down view of a power semiconductor module comprised in the power electronic system (FIG. 2B).

[0009] FIGS. 3A and 3B show a top down view (FIG. 3A) and a side view (FIG. 3B) of a further power semiconductor module.

[0010] FIG. 4 shows a sectional view of the power semiconductor module of FIGS. 1A and 1B according to a specific example.

[0011] FIG. 5 is a flow chart of an exemplary method for fabricating a power semiconductor module.

DETAILED DESCRIPTION

[0012] In the following detailed description, known structures and elements are shown in schematic form in order to facilitate describing one or more aspects of the disclosure. In this regard, directional terminology, such as “top”, “bottom”, “left”, “right”, “upper”, “lower” etc., is used with reference to the orientation of the Figure(s) being described. Because components of the disclosure can be positioned in a number of different orientations, the directional terminology is used for purposes of illustration only. It is to be understood that other examples may be utilized and structural or logical changes may be made.

[0013] In addition, while a particular feature or aspect of an example may be disclosed with respect to only one of several implementations, such feature or aspect may be combined with one or more other features or aspects of the other implementations as may be desired and advantageous for any given or particular application, unless specifically noted otherwise or unless technically restricted. Furthermore, to the extent that the terms “include”, “have”, “with” or other variants thereof are used in either the detailed description or the claims, such terms are intended to be inclusive in a manner similar to the term “comprise”. The terms “coupled” and “connected”, along with derivatives thereof may be used. It should be understood that these terms may be used to indicate that two elements cooperate or interact with each other regardless whether they are in direct physical or electrical contact, or they are not in direct contact with each other; intervening elements or layers may be provided between the “bonded”, “attached”, or “connected” elements. However, it is also possible that the “bonded”, “attached”, or “connected” elements are in direct contact with each other. Also, the term “exemplary” is merely meant as an example, rather than the best or

optimal.

[0014] The semiconductor die(s) of a power semiconductor module may be covered with an electrically insulating encapsulation material in order to be embedded in an encapsulation. The encapsulation material may, for example, comprise or consist of any appropriate plastic or polymer material and may e.g. contain inorganic filler materials configured to reduce the thermal resistance of the encapsulation. Various techniques may be employed to encapsulate the semiconductor die(s) with the encapsulation material, for example compression molding, injection molding, transfer molding, or lamination.

[0015] An efficient power semiconductor module, an efficient power electronic system and an efficient method for fabricating a power semiconductor module may for example reduce material consumption, ohmic losses, chemical waste, etc. and may thus enable energy and/or resource savings. Improved power semiconductor modules, improved power electronic systems and improved methods for fabricating a power semiconductor module, as specified in this description, may thus at least indirectly contribute to green technology solutions, i.e. climate-friendly solutions providing a mitigation of energy and/or resource use.

[0016] FIGS. **1A** and **1B** show a power semiconductor module **100** comprising an encapsulation body **110**, a first power lead **120**, a second power lead **130**, a plurality of power semiconductor dies **140**, a first pin **150** and a second pin **160**. FIG. **1A** shows a top down view and FIG. **1B** shows a side view of the power semiconductor module **100**. The power semiconductor dies **140** are omitted from FIG. **1A** and are shown with dashed lines in FIG. **1B** since the encapsulation body **110** obstructs the view onto the power semiconductor dies **140**.

[0017] The power semiconductor module **100** may comprise any suitable electrical circuit, for example a converter circuit, an inverter circuit, a half bridge circuit, a full bridge circuit, etc. The power semiconductor module **100** may be configured to operate with a high electrical voltage and or a strong electrical current, for example a voltage of 100V or more, or 500V or more, or 1 kV or more and/or a current of 1 A or more, or 10 A or more. The power semiconductor module **100** may for example be configured for use in automotive applications.

[0018] The encapsulation body **110** comprises a first side **111**, an opposite second side **112** and lateral sides **113** connecting the first and second sides **111**, **112**. The lateral sides **113** may for example have a smaller surface area than the first and second sides **111**, **112**. The first and second sides **111**, **112** may for example have about the same surface area or exactly the same surface area.

[0019] The power semiconductor module **100** may have any suitable shape and any suitable dimensions. For example, the power semiconductor module **100** may have an essentially rectangular or quadratic shape as viewed from above the first side **111**. The first side **111** may for example have an edge length of 1 cm or more, or 5 cm or more, or 10 cm or more, or 15 cm or more.

[0020] According to an example, the encapsulation body **110** comprises or consists of a molded body. Such a molded body may for example be fabricated using a technique like compression molding, injection molding or transfer molding. According to another example, the encapsulation body **110** comprises a plastic frame defining an interior volume, wherein the power semiconductor dies **140** are arranged within the interior volume. A potting material may be deposited on the power semiconductor dies **140** and may at least partially fill the interior volume.

[0021] The power semiconductor module **100** may for example be configured for single side cooling, meaning that a single side of the encapsulation body **110** is configured to be coupled to a heatsink. For example, the second side **112** of the encapsulation body **110** may be configured to be coupled to a heatsink. The first side **111** may for example be configured to face a driver board. According to another example, the power semiconductor module **100** may be configured for double sided cooling, wherein both the first and second sides **111**, **112** are configured to be coupled to heatsinks. In this case, a driver board may e.g. be arranged above such a heatsink.

[0022] The first power lead **120** and the second power lead **130** are exposed from a first one of the

lateral sides **113** of the encapsulation body **110**. The first and second power leads **120, 130** may be arranged side-by-side. The first and second power leads **120, 130** may be arranged coplanar. However, it is also possible that the first and second power leads **120, 130** are arranged in different planes relative to each other.

[0023] The first and second power leads **120, 130** may comprise or consist of any suitable metal or metal alloy. The first and second power leads **120, 130** may for example comprise or consist of Al or Cu. According to an example, the first and second power leads **120, 130** are coated with a suitable plating, e.g. a Ni plating. According to an example, the first and second power leads **120, 130** are leadframe parts.

[0024] The first power lead **120** is configured to provide a first DC+ connection (i.e. a direct current connection with a positive voltage) for the power semiconductor module **100** (in particular, for the power semiconductor dies **140**) and the second power lead **130** is configured to provide a first DC- connection (i.e. a direct current connection with a negative voltage) for the power semiconductor module **100**.

[0025] According to an example, the power semiconductor module **100** may comprise a further power lead **170** which may for example be configured to provide a phase connection for the power semiconductor module **100**. The further power lead **170** may comprise or consist of the same material as the first and second power leads **120, 130** and may also be a leadframe part. The further power lead **170** may for example be exposed from a second one of the lateral sides **113** of the encapsulation body **110**. The second lateral side **113** may for example be opposite to the first lateral side **113** comprising the first and second power leads **120, 130**.

[0026] According to an example, the power semiconductor module **100** may comprise even further power leads. For example, the power semiconductor module **100** may comprise one or more further power leads which are configured to be part of the first DC+ connection and/or one or more further power leads which are configured to be part of the first DC- connection and/or one or more further power leads which are configured to be part of the phase connection.

[0027] The power semiconductor dies **140** are encapsulated by the encapsulation body **110** and are electrically connected to the first and second power leads **120, 130**. Although the power semiconductor module **100** so far has been described as having at least two power semiconductor dies **140**, it is of course also possible that the power semiconductor module **100** comprises e.g. two power transistor circuits provided in monolithic integration.

[0028] The power semiconductor dies **140** may be arranged over one or more carriers (not shown). For example, the power semiconductor dies **140** may be soldered or sintered or glued with conductive glue to the carrier(s). The power leads **120, 130, 170** and possibly the first and second pins **150, 160** and possibly the further pins **180** (see below) may also be arranged over the carrier(s). The power semiconductor dies **140** and the power leads **120, 130, 170** may for example be electrically connected via the carrier(s).

[0029] The carrier(s) may for example be power electronic substrates, e.g. substrates of the type direct copper bonded (DCB), direct aluminum bonded (DAB), active metal brazed (AMB), insulated metal substrate (IMS), etc. The carrier(s) may for example be exposed from the second side **112** of the encapsulation **110**. In the case that the power semiconductor module **100** is configured for double sided cooling, a further carrier may be exposed from the first side **111**.

[0030] According to an example, the power semiconductor module **100** is configured for particular fast switching of a load current. To this end, the power semiconductor module **100** may e.g. comprise power semiconductor dies **140** comprising or consisting of SiC or GaN. Due to this fast switching, it may be beneficial to address parasitic inductances in the power semiconductor module **100** as outlined further below.

[0031] The first and second pins **150, 160** are exposed from the encapsulation body **110**. In the example shown in FIG. 1, the first and second pins **150, 160** are exposed from the first side **111** of the encapsulation body **110**. It is, however, also possible that the first and second pins **150, 160** are

e.g. exposed from one or more of the lateral sides **113**.

[0032] The first and second pins **150, 160** are arranged perpendicular to the first and second power leads **120, 130**. This may mean that the whole of the first and second pins **150, 160** or at least an upper end of the first and second pins **150, 160** points in a direction that is perpendicular to the first and second power leads **120, 130**. The first and second pins **150, 160** may be configured to be connected to an external appliance (e.g. a driver board), wherein the external appliance may e.g. be arranged above the first side **111** of the encapsulation body **110** (this may be the reason why the first and second pins **150, 160** are arranged perpendicular to the first and second power leads **120, 130**).

[0033] According to an example, the first and second pins **150, 160** are press-fit pins, configured to provide a press-fit connection with an external appliance. The first and second pins **150, 160** may comprise or consist of any suitable metal or metal alloy, e.g. Al or Cu. The first and second pins **150, 160** may comprise a suitable plating, e.g. a Ni plating.

[0034] The first pin **150** is configured to provide a second DC+ connection and the second pin **160** is configured to provide a second DC- connection of the power semiconductor module **100**. This may in particular mean that the first pin **150** is electrically connected to the first power lead **120** and is at the same electrical potential as the first power lead **120** and the second pin **160** is electrically connected to the second power lead **130** and is at the same electrical potential as the second power lead **130**.

[0035] According to an example, an external appliance like a driver board may be arranged over the first side **111** of the encapsulation body **110**, wherein the external appliance comprises a first capacitor (e.g. a fast ceramic capacitor) which is connected to the first and second pins **150, 160** to provide a first commutation loop. Providing such a commutation loop, wherein the DC+ potential and DC- potential of the power leads **120, 130** are connected to the first capacitor, may e.g. help with reducing current overshoots. Due to a lower inductance of this commutation loop compared to the connection between the power leads **120, 130** and an external DC link capacitor, the current can commute more quickly and overshoots between Drain-Source-voltages in the power semiconductor module **100** may be reduced.

[0036] Since the first and second pins **150, 160** may solely be configured to provide this commutation loop, the first and second pins **150, 160** need not necessarily be configured to carry a load current. Compared to the first and second power leads **120, 130**, the first and second pins **150, 160** may therefore e.g. have a smaller cross section. For example, the cross section of the first and second pins **150, 160** may be less than half or less than one quarter of the cross section of the first and second power leads **120, 130**.

[0037] According to an example, the power semiconductor module **100** may comprise one or more further pins **180** exposed from the encapsulation body **110**. The one or more further pins **180** may for example be configured to transmit control signals for driving the power semiconductor dies **140**. Additionally or alternatively, the one or more further pins **180** may be configured to transmit sensing signals, e.g. drain voltage sensing or source voltage sensing signals.

[0038] FIG. 2A shows a perspective view of a power electronic system **200** comprising a driver board **210** and a power semiconductor module **220**. FIG. 2B shows a top down view of the power semiconductor module **220**. The power semiconductor module **220** may be similar or identical to the power semiconductor module **100**, except for the differences described in the following.

[0039] The power semiconductor module **220** comprises the first and second power leads **120, 130** and the power semiconductor module **220** additionally comprises a third power lead **222**. The third power lead **222** may for example be configured to be part of the first DC+ connection, parallel to the first power lead **120** (this example is shown in FIG. 2A). According to another example, it is the other way around, meaning that the first and third power leads **120, 222** provide the first DC- connection and the second power lead **130** provides the first DC+ connection of the power semiconductor module **220**.

[0040] The power semiconductor module **220** further comprises the first and second pins **150**, **160** and the power semiconductor module **220** additionally comprises a third pin **224** and a fourth pin **226**. The first pin **150** may be connected in parallel with the first power lead **120**, the second and fourth pins **160**, **226** may be connected in parallel with the second power lead **130** and the third pin **224** may be connected in parallel with the third power lead **222**. The first and third pins **150**, **224** may be configured to provide the second DC+ connection and the second and fourth pins **160**, **226** may be configured to provide the second DC- connection (this example is shown in FIG. 2A). According to another example, it is the other way around.

[0041] The driver board **210** may for example comprise a printed circuit board (PCB). The driver board **210** comprises driver circuitry **212** configured to drive the power semiconductor dies of the power semiconductor module **220**. The driver circuitry **212** may in particular be connected to gate electrodes of the power semiconductor dies of the power semiconductor module **220**. The driver circuitry **212** may be connected to the power semiconductor module **220** via one or more further pins **180**.

[0042] The driver board **210** further comprises a first capacitor **214** connected to the first and second pins **150**, **160** to provide a first commutation loop and a second capacitor **216** connected to the third and fourth pins **224**, **226** to provide a second commutation loop. In FIG. 2A, the first and second commutation loops are indicated by dashed lines. The capacitors **214**, **216** may for example be fast ceramic capacitors. The first, second and third power leads **120**, **130**, **222** may be configured to be connected to an external DC link capacitor, e.g. via welding, soldering or screwing. Such a DC link capacitor may for example be a foil winded capacitor and may be comparatively large. Furthermore, such a DC link capacitor may have a comparatively high capacitance. The first and second capacitors **214**, **216** may have a smaller capacitance than the DC link capacitor and/or may be faster than the DC link capacitor.

[0043] According to an example, the first and second capacitors **214**, **216** each have a capacitance in the range of about 100 nF to about 1 μ F. The lower limit of this range may also be about 200 nF or about 400 nF and the upper limit may also be about 800 nF or about 600 nF. A DC link capacitor on the other hand may e.g. have a capacitance in the range of 100 μ F to 1 mF. In other words, the capacitances of the first and second capacitors **214**, **216** may be a factor of 100 or more or a factor of 1000 or more smaller than a capacitance of a dedicated DC link capacitor.

[0044] FIGS. 3A and 3B show a further power semiconductor module **300** which may be similar or identical to the power semiconductor modules **100** and **220**, except for the differences described in the following. FIG. 3A shows a top down view of the power semiconductor module **300** and FIG. 3B shows a side view along the arrow B in FIG. 3A.

[0045] In the power semiconductor module **300**, the first pin **150** and the second pin **160** are exposed from one or more of the lateral sides **113** of the encapsulation body **110**. In particular, the first and second pins **150**, **160** may be exposed from opposite ones of the lateral sides **113** (compare FIG. 3A). Furthermore, the first and second pins **150**, **160** may each comprise a first portion arranged in a first plane and a second portion arranged in a different second plane, wherein the first and second planes are arranged at a non-zero angle relative to each other. The non-zero angle may for example be about 90°. The first and second power leads **120**, **130** may for example be arranged in the first plane. In other words, the first and second pins **150**, **160** may be bent upwards such that the second portion of the first and second pins is arranged perpendicular to the first and second power leads **120**, **130**.

[0046] The power semiconductor module **300** may also comprise the one or more further pins **180**. The one or more further pins **180** may also be exposed from one or more of the lateral sides **113** of the encapsulation body **110** (for example, the one or more further pins **180** may be exposed from the same lateral sides **113** as the first and second pins **120**, **130**, compare FIG. 3A).

[0047] The power semiconductor module **300** may for example be configured for double sided cooling. This may in particular mean that a first carrier (e.g. a DCB) is exposed from the first side

111 of the encapsulation body **110** and a second carrier (e.g. a DCB) is exposed from the second side **112** of the encapsulation body **110**. This may be the reason why the pins **150**, **160**, **180** are exposed from the lateral sides **113** of the encapsulation body **110**.

[0048] FIG. 4 shows a sectional view of the power semiconductor module **100** according to a specific example.

[0049] As shown in FIG. 4, the first pin **150** may be in direct contact with the first power lead **120**. In a similar manner, the second pin **160** may be in direct contact with the second power lead **130**. This may for example mean that lower ends of the first and second pins **150**, **160** are inserted into respective recesses **190** in the first and second power leads **120**, **130**.

[0050] According to an example, the first and second power leads **120**, **130** are metal clips. The recesses **190** may be fabricated in the metal clips using any suitable process, e.g. a punching process or a drilling process. The first and second pins **150**, **160** and the recesses **190** may for example form press-fit connections.

[0051] As shown in FIG. 4, the lower ends of the first and second pins **150**, **160** may be tapered in order to make inserting the pins **150**, **160** into the recesses **190** easier. Upper ends of the pins **150**, **160** may also be tapered in order to make inserting the pins **150**, **160** into through holes of a driver board like the driver board **210** easier.

[0052] According to an example, the first and second power leads **120**, **130** (and possibly also the further power lead **170**) and the power semiconductor die(s) **140** are arranged on a carrier **192**. The power leads **120**, **130** (and **170**) and the power semiconductor die(s) **140** may for example be soldered or sintered or glued with conductive glue onto the carrier **192**. The carrier **192** may for example be a power electronic substrate, e.g. a power electronic substrate of one the types mentioned further above.

[0053] FIG. 5 is a flow chart of a method **500** for fabricating a power semiconductor module. The method **500** may for example be used to fabricate the power semiconductor modules **100**, **220** or **300**.

[0054] The method **500** comprises at **501** a process of connecting a plurality of power semiconductor dies to a first and second power lead, the first power lead configured to provide a first DC+ connection and the second power lead configured to provide a first DC- connection; at **502** a process of encapsulating the plurality of power semiconductor dies with an encapsulation body, the encapsulation body comprising a first side, an opposite second side and lateral sides connecting the first and second sides, such that the first and second power leads are exposed from a first one of the lateral sides; and at **503** a process of providing a first and second pin exposed from the encapsulation body and arranged perpendicular to the first and second power leads, the first pin configured to provide a second DC+ connection and the second pin configured to provide a second DC- connection.

[0055] According to an example of the method **500**, the first and second pins **150**, **160** are provided after the power semiconductor dies **140** have been encapsulated. This may entail fabricating (e.g. by molding) the encapsulation body **110** such that openings for the pins are provided. Alternatively, such openings may be fabricated after the encapsulation body **110** has been fabricated by removing part of the encapsulation **110** body by e.g. drilling, cutting, grinding, etc.

[0056] In the following, the power semiconductor module, the power electronic system and the method for fabricating a power electronic system are explained using specific examples.

[0057] Example 1 is a power electronic system, comprising: a power semiconductor module, comprising: an encapsulation body comprising a first side, an opposite second side and lateral sides connecting the first and second sides, a first and a second power lead exposed from a first one of the lateral sides, the first power lead configured to provide a first DC+ connection and the second power lead configured to provide a first DC- connection, a plurality of power semiconductor dies encapsulated by the encapsulation body and connected to the first and second power leads, a first and a second pin exposed from the encapsulation body and arranged perpendicular to the first and

second power leads, the first pin configured to provide a second DC+ connection and the second pin configured to provide a second DC- connection; and a driver board arranged over the first side of the encapsulation body and comprising driver circuitry configured to drive the power semiconductor dies of the power semiconductor module, wherein the driver board further comprises a first capacitor connected to the first and second pins to provide a first commutation loop.

[0058] Example 2 is the power electronic system of example 1, wherein the first and second pins are press-fit pins.

[0059] Example 3 is the power electronic system of example 1 or 2, wherein the first and second power leads are connected to a direct current link capacitor via soldered joints, welded joints or joints comprising screws.

[0060] Example 4 is the power electronic system of one of the preceding examples, wherein the first capacitor has a capacitance in the range of 50 nF to 2000 nF.

[0061] Example 5 is the power electronic system of one of the preceding examples, wherein the first and second power leads are metal clips and wherein lower ends of the first and second pins are in direct contact with the metal clips.

[0062] Example 6 is the power electronic system of example 5, wherein the lower ends of the first and second pins are inserted into recesses in the metal clips.

[0063] Example 7 is the power electronic system of one of the preceding examples, wherein the first and second pins are exposed from the first side of the encapsulation body.

[0064] Example 8 is the power electronic system of one of examples 1 to 6, wherein the first and second pins are exposed from one or more of the lateral sides of the encapsulation body and wherein the first and second pins are bent upwards to be perpendicular to the first and second power leads.

[0065] Example 9 is the power electronic system of one of the preceding examples, wherein the encapsulation body is a molded body.

[0066] Example 10 is the power electronic system of one of examples 1 to 8, wherein the encapsulation body comprises a plastic frame, wherein the power semiconductor dies are arranged within an interior volume defined by the plastic frame.

[0067] Example 11 is the power electronic system of one of the preceding examples, wherein a load current flows through the first and second power leads but not through the first and second pins.

[0068] Example 12 is a power semiconductor module, comprising: an encapsulation body comprising a first side, an opposite second side and lateral sides connecting the first and second sides; a first and a second power lead exposed from a first one of the lateral sides, the first power lead configured to provide a first DC+ connection and the second power lead configured to provide a first DC- connection; a plurality of power semiconductor dies encapsulated by the encapsulation body and connected to the first and second power leads; and a first and a second pin exposed from the encapsulation body and arranged perpendicular to the first and second power leads, the first pin configured to provide a second DC+ connection and the second pin configured to provide a second DC- connection.

[0069] Example 13 is the power semiconductor module of example 12, wherein the first and second pins are press-fit pins.

[0070] Example 14 is the power semiconductor module of example 12 or 13, wherein the first and second power leads are configured to be connected to an external appliance via soldered joints, welded joints or joints comprising screws.

[0071] Example 15 is the power semiconductor module of one of examples 12 to 14, wherein the power semiconductor dies are connected to form a half bridge circuit.

[0072] Example 16 is a method for fabricating a power semiconductor module, the method comprising: connecting a plurality of power semiconductor dies to a first and second power lead,

the first power lead configured to provide a first DC+ connection and the second power lead configured to provide a first DC- connection; encapsulating the plurality of power semiconductor dies with an encapsulation body, the encapsulation body comprising a first side, an opposite second side and lateral sides connecting the first and second sides, such that the first and second power leads are exposed from a first one of the lateral sides; and providing a first and second pin exposed from the encapsulation body and arranged perpendicular to the first and second power leads, the first pin configured to provide a second DC+ connection and the second pin configured to provide a second DC- connection.

[0073] Example 17 is the method of example 16, wherein the first and second pins are provided after the power semiconductor dies have been encapsulated.

[0074] Example 18 is the method of example 16 or 17, wherein providing the first and second pins comprises inserting lower ends of the first and second pins into recesses in the first and second power leads.

[0075] Example 19 is an apparatus comprising means for performing the method according to anyone of examples 16 to 18.

[0076] While the disclosure has been illustrated and described with respect to one or more implementations, alterations and/or modifications may be made to the illustrated examples without departing from the spirit and scope of the appended claims. In particular regard to the various functions performed by the above described components or structures (assemblies, devices, circuits, systems, etc.), the terms (including a reference to a “means”) used to describe such components are intended to correspond, unless otherwise indicated, to any component or structure which performs the specified function of the described component (e.g., that is functionally equivalent), even though not structurally equivalent to the disclosed structure which performs the function in the herein illustrated exemplary implementations of the disclosure.

Claims

1. A power electronic system, comprising: a power semiconductor module, comprising: an encapsulation body comprising a first side, an opposite second side, and lateral sides connecting the first and second sides; a first and a second power lead exposed from a first one of the lateral sides, the first power lead configured to provide a first DC+ connection and the second power lead configured to provide a first DC- connection; a plurality of power semiconductor dies encapsulated by the encapsulation body and connected to the first and second power leads; a first and a second pin exposed from the encapsulation body and arranged perpendicular to the first and second power leads, the first pin configured to provide a second DC+ connection and the second pin configured to provide a second DC- connection; and a driver board arranged over the first side of the encapsulation body and comprising driver circuitry configured to drive the power semiconductor dies of the power semiconductor module, wherein the driver board further comprises a first capacitor connected to the first and second pins to provide a first commutation loop.
2. The power electronic system of claim 1, wherein the first and second pins are press-fit pins.
3. The power electronic system of claim 1, wherein the first and second power leads are connected to a direct current link capacitor via soldered joints, welded joints or joints comprising screws.
4. The power electronic system of claim 1, wherein the first capacitor has a capacitance in a range of 50 nF to 2000 nF.
5. The power electronic system of claim 1, wherein the first and second power leads are metal clips, and wherein lower ends of the first and second pins are in direct contact with the metal clips.
6. The power electronic system of claim 5, wherein the lower ends of the first and second pins are inserted into recesses in the metal clips.
7. The power electronic system of claim 1, wherein the first and second pins are exposed from the first side of the encapsulation body.

- 8.** The power electronic system of claim 1, wherein the first and second pins are exposed from one or more of the lateral sides of the encapsulation body, and wherein the first and second pins are bent upwards to be perpendicular to the first and second power leads.
 - 9.** The power electronic system of claim 1, wherein the encapsulation body is a molded body.
 - 10.** The power electronic system of claim 1, wherein the encapsulation body comprises a plastic frame, and wherein the power semiconductor dies are arranged within an interior volume defined by the plastic frame.
 - 11.** The power electronic system of claim 1, wherein a load current flows through the first and second power leads but not through the first and second pins.
 - 12.** A power semiconductor module, comprising: an encapsulation body comprising a first side, an opposite second side, and lateral sides connecting the first and second sides; a first and a second power lead exposed from a first one of the lateral sides, the first power lead configured to provide a first DC+ connection and the second power lead configured to provide a first DC− connection; a plurality of power semiconductor dies encapsulated by the encapsulation body and connected to the first and second power leads; and a first and a second pin exposed from the encapsulation body and arranged perpendicular to the first and second power leads, the first pin configured to provide a second DC+ connection and the second pin configured to provide a second DC− connection.
 - 13.** The power semiconductor module of claim 12, wherein the first and second pins are press-fit pins.
 - 14.** The power semiconductor module of claim 12, wherein the first and second power leads are configured to be connected to an external appliance via soldered joints, welded joints or joints comprising screws.
 - 15.** The power semiconductor module of claim 12, wherein the power semiconductor dies are connected to form a half bridge circuit.
 - 16.** A method for fabricating a power semiconductor module, the method comprising: connecting a plurality of power semiconductor dies to a first and second power lead, the first power lead configured to provide a first DC+ connection and the second power lead configured to provide a first DC− connection; encapsulating the plurality of power semiconductor dies with an encapsulation body, the encapsulation body comprising a first side, an opposite second side, and lateral sides connecting the first and second sides, such that the first and second power leads are exposed from a first one of the lateral sides; and providing a first and a second pin exposed from the encapsulation body and arranged perpendicular to the first and second power leads, the first pin configured to provide a second DC+ connection and the second pin configured to provide a second DC− connection.
 - 17.** The method of claim 16, wherein the first and second pins are provided after the power semiconductor dies have been encapsulated.
 - 18.** The method of claim 16, wherein providing the first and second pins comprises inserting lower ends of the first and second pins into recesses in the first and second power leads
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